

AYESHA SIDDIKHA SYED

Data Engineer

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Data Engineer with 3+ years of experience designing, building, and optimizing scalable data pipelines and infrastructure across financial and credit risk domains. Applies modern data stack technologies including Azure Synapse, Snowflake, Databricks, Kafka, and AWS Glue to manage high-volume batch and real-time data workflows. Implements secure and compliant data solutions aligned with GDPR and CCPA to enhance system reliability and support data-driven decision-making. Achieves measurable improvements by optimizing data processing efficiency, automating ETL workflows, and delivering actionable insights through Power BI and Tableau.

PROFESSIONAL EXPERIENCE

Data Engineer | Northern Trust

Nov 2024 – Present

- Architected secure data pipelines for financial transactions using Azure Data Factory and Azure Synapse, ensuring end-to-end encryption and reducing data transfer failures by 90%.
- Integrated external market feeds and custodial data from over 10 financial partners, enabling real-time NAV calculations and reducing reconciliation time by 35%.
- Implemented delta lake architecture using Databricks, supporting scalable data lake operations and accelerating large dataset queries by 50%.
- Collaborated with risk and compliance teams to build audit-ready data lineage tracking, ensuring 100% traceability of sensitive financial data for regulatory reviews.
- Optimized Python-based transformation scripts to improve runtime by 40%, supporting daily batch jobs that handle over 2 TB of financial data.
- Developed anomaly detection mechanisms using machine learning models in PySpark to flag irregular fund activity, reducing manual review time by 60%.
- Created financial reporting datasets in Power BI-compatible formats, reducing dashboard refresh latency by 45% and increasing stakeholder access to near real-time insights.

Data Engineer | Equifax, USA

May 2024 – Oct 2024

- Led the migration of on-premise data warehouses to Snowflake on AWS, reducing infrastructure costs by 35% and improving query performance by 60%.
- Engineered batch and streaming data solutions using Kafka and AWS Glue, enabling real-time credit risk analysis and reducing processing delays by 45%.
- Developed and maintained over 20 ELT pipelines using SQL, Python, and dbt, ensuring high data availability and consistent delivery for critical business metrics.
- Collaborated with cross-functional teams to implement GDPR and CCPA compliance controls across all data processes, resulting in 100% adherence during internal audits.
- Performed advanced data modeling and partitioning strategies in Snowflake and Redshift, which improved report generation speeds by 70% for enterprise-wide dashboards.
- Automated schema evolution and version control using GitHub Actions and Terraform, streamlining CI/CD integration and reducing deployment time by 40%.

Data Engineer | Infosys

May 2021 – Dec 2022

- Designed and deployed scalable data pipelines using Apache Spark and Python to process structured and unstructured data, improving ETL efficiency by 30%.
- Optimized data workflows on AWS (S3, EMR, Redshift) and Azure Data Factory, reducing data latency by 40% and enabling near real-time analytics.
- Collaborated with data scientists and analysts to integrate business logic into data models, improving data usability and reducing report turnaround time by 25%.
- Performed data ingestion and transformation from diverse sources including RDBMS, flat files, and APIs, ensuring data quality and consistency across environments.
- Implemented data quality checks and monitoring using PySpark and custom Python scripts, proactively identifying and resolving anomalies to maintain data integrity.

EDUCATION

- MS in **Computer Information Systems**, University of Central Missouri, USA
- BS in **Mechanical Engineering**, Jawaharlal Nehru Technological University, India

Jan 2023 - Aug 2024

Mar 2016 - Sept 2020

TECHNICAL SKILLS

- **Programming & Scripting:** Python, SQL, PySpark, Scala, Shell Scripting
- **Data Engineering & ETL Tools:** Apache Spark, Apache Kafka, AWS Glue, dbt, Informatica, Azure Data Factory
- **Cloud Platforms:** AWS (S3, Redshift, EMR, Lambda), Azure (Synapse, Blob Storage, ADF), Databricks
- **Data Warehousing & Databases:** Snowflake, Amazon Redshift, Microsoft SQL Server, PostgreSQL, MySQL
- **Big Data & Lakehouse Technologies:** Delta Lake, Hadoop, HDFS
- **Data Modeling & Orchestration:** Star/Snowflake Schema, dbt, Airflow, GitHub Actions, Terraform
- **Data Visualization & BI Tools:** Power BI, Tableau
- **DevOps & CI/CD:** Git, GitHub Actions, Terraform, Jenkins, Docker
- **Compliance & Data Governance:** GDPR, CCPA, Data Lineage, Audit Logging
- **Machine Learning & Analytics:** scikit-learn, Pandas, NumPy, Matplotlib